

Enterprise Information Management

Data Standards

Enterprise Data Management and Governance Standards

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SENSITIVITY CLASSIFICATION: Internal Use

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Revision History

Version	Date	Changed By	Description
1.0	2024-06-14	R. Okonjo	Initial publication
2.0	2025-11-02	R. Okonjo	Added lifecycle and disposal
2.1	2026-04-01	M. Ferreira	Classification tiers revised

1. Introduction

1.1 Purpose

This document defines the Data Standards that govern how data is created, structured, stored, secured, integrated and retired across Northwind Energy. It is technology-agnostic and applies regardless of the platform or vendor delivering a solution.

1.2 Scope

These standards apply to all structured, semi-structured and unstructured data created, processed, stored or transmitted by enterprise systems. They are binding for internal teams, contractors and external vendors delivering data-related solutions on behalf of the organisation.

1.3 Definitions of Terms

Term	Definition
Data Owner	The role accountable for the definition, quality and appropriate use of a data domain.
Data Steward	The role responsible for day-to-day application of data standards within a domain.
Golden Record	The single trusted version of a data entity after reconciliation across sources.
PII	Personally Identifiable Information · data that can identify an individual directly or indirectly.

2. Data Governance Standards

2.1 Data Ownership and Stewardship

Statement: Every data domain must have a designated Data Owner accountable for its definition and quality, supported by one or more Data Stewards.

Rationale: Clear ownership prevents ambiguity in decision-making and provides an escalation path when quality issues arise.

Requirements:

- Every critical data domain must have a named Data Owner recorded in the data catalogue
- Data Owners must approve changes to data definitions and classification within their domain
- Stewardship responsibilities must be reviewed at least annually

2.2 Policy Compliance and Monitoring

Statement: Compliance with these standards must be actively monitored and reported, with corrective action tracked to closure.

Rationale: Without active monitoring, governance policies become documentation exercises rather than enforced controls.

Requirements:

- Non-compliant systems must be logged in a governance risk register with a remediation plan
- Key governance metrics must be reported to the Data Governance Council quarterly

3. Data Architecture Standards

3.1 Layered Data Architecture

Statement: Data platforms must implement a layered architecture separating raw, curated and consumption-ready data.

Rationale: Layering isolates change and allows reprocessing without impacting downstream consumers.

Requirements:

- Raw data must be retained in its original form prior to transformation
- Data must not be modified in place within the raw layer
- Consumption layers must be modelled for their specific reporting or operational use

3.2 Single Source of Truth

Statement: Each data entity must have one authoritative system designated as its source of truth, from which all consuming systems derive their copy.

Rationale: Multiple uncontrolled sources of the same data lead to conflicting values and loss of trust in reporting.

Requirements:

- Downstream systems must synchronise from the system of record rather than re-entering data
- Direct database-to-database connections into a system of record are prohibited; access must be mediated by the integration platform
- Any temporary duplication must be clearly labelled as a derived copy

4. Data Modelling Standards

4.1 Naming Conventions

Statement: All database objects must follow the documented naming convention below.

Rationale: Consistent naming reduces ambiguity and supports automated tooling such as catalogues and code generation.

Requirements:

- Names must be descriptive, in English, and use approved abbreviations only
- Reserved words and platform-specific keywords must be avoided
- Conventions must be applied consistently across dev, test and production

Object Type	Convention	Example
Table	snake_case, plural noun	customer_orders
Column	snake_case, singular noun	order_date
Primary Key	pk_<table>	pk_customer_orders
Foreign Key	fk_<table>_<ref_table>	fk_orders_customers
Index	ix_<table>_<columns>	ix_orders_order_date
View	vw_<business_name>	vw_active_customers

4.2 Standard Data Types and Formats

Statement: Common data domains such as dates, currency and identifiers must use standardised types and formats across all systems.

Rationale: Inconsistent formats are a leading cause of integration defects and reporting errors.

Requirements:

- Dates and timestamps must use ISO 8601 for system-to-system exchange
- Monetary values must specify a currency code and a consistent decimal precision
- Character encoding must default to UTF-8 across all stores and interfaces

5. Data Quality Standards

5.1 Data Quality Dimensions

Dimension	Definition	Example Metric
Accuracy	Data correctly reflects the real-world object it represents	% matching verified source
Completeness	Required attributes are populated	% mandatory fields populated
Timeliness	Data is available within the required window	Average latency vs SLA
Uniqueness	No unintended duplicates exist for the same entity	Duplicate record rate

5.2 Quality Monitoring and Remediation

Statement: Critical data domains must be subject to automated quality monitoring with defined thresholds and remediation workflows.

Rationale: Proactive monitoring surfaces issues before they propagate into downstream reporting and operational decisions.

Requirements:

- Automated checks must run on ingestion and on a scheduled interval
- Breaches of threshold must raise an alert and be logged for remediation
- Root-cause analysis must be performed for recurring issues

6. Data Security and Privacy Standards

6.1 Data Classification

All data assets must be classified according to sensitivity to determine the appropriate handling, access and protection controls. Every solution design must state the classification of the data it holds.

Classification	Default Access	Encryption at Rest	External Sharing
Public	Everyone	Not required	Permitted
Internal	All employees	Recommended	Requires approval
Confidential	Need-to-know	Required	Contractual controls
Restricted	Named roles	Required	Prohibited

6.2 Access Control

Statement: Access to data must be granted on the principle of least privilege using role-based access control aligned to data classification.

Rationale: Least privilege limits the blast radius of a compromised credential and supports auditability of who can access what.

Requirements:

- Access must be requested, approved and recorded
- Access must be re-certified at least annually for Confidential and Restricted data
- Privileged and administrative access must be logged and monitored

6.3 Encryption

Statement: Confidential and Restricted data must be encrypted both at rest and in transit using currently approved algorithms.

Rationale: Encryption protects confidentiality even when the underlying storage or network is compromised.

Requirements:

- Data in transit must use TLS 1.2 or higher, or an equivalent current standard
- Deprecated protocols and weak cipher suites must be disabled
- Encryption keys must be managed separately from the data they protect

7. Data Integration Standards

7.1 Integration Pattern Selection

Statement: Integration must use an approved pattern · batch, asynchronous messaging, event streaming or API · appropriate to the latency and volume of the use case.

Rationale: Standardised patterns reduce point-to-point complexity and make the integration landscape easier to monitor and maintain.

Requirements:

- Point-to-point integrations must be avoided in favour of the shared integration platform
- Real-time integration must be used where near-immediate availability is required
- Batch integration must have clearly defined windows and dependencies

7.2 Resilience and Idempotency

Statement: Integrations must handle failure gracefully through retries, dead-letter handling and idempotent processing.

Rationale: Network and system failures are inevitable; resilient design ensures data is not lost, duplicated or corrupted when they occur.

Requirements:

- Failed messages that cannot be processed after retries must be routed to a dead-letter queue
- Integrations must be idempotent so repeated delivery does not create duplicate outcomes
- Each transaction must carry a unique correlation identifier for traceability

8. Data Lifecycle Management Standards

8.1 Retention and Archival

Statement: All data must be retained and archived according to the periods defined below, based on legal, regulatory and business requirements.

Rationale: Structured retention ensures long-term integrity and regulatory compliance while managing storage cost.

Requirements:

- Retention periods must be defined per data category and recorded in the catalogue
- Backup and archival processes must align with the defined retention period
- Auditability must be preserved throughout the lifecycle

Retention tiers. Tier 4 periods are set per system and are not fixed centrally.

Tier	Category	Retention Period	Archive After	Disposal Method
Tier 1	Regulated financial records	7 years	2 years	Certified destruction
Tier 2	Customer contract records	6 years	2 years	Certified destruction
Tier 3	Operational telemetry	90 days	30 days	Automated purge

Tier 4	Transient diagnostic logs			Automated purge
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8.2 Secure Disposal

Statement: Data must be securely and irreversibly disposed of at the end of its lifecycle in accordance with retention, legal and regulatory requirements.

Rationale: Ensures expired data is permanently removed while maintaining auditability.

Requirements:

- Disposal must be auditable, authorised and verifiable
- All copies and dependent systems must be identified and addressed before destruction
- Legal holds must override automated disposal

9. Records Management Standards

This section is under development.

10. Roles and Responsibilities

The following matrix summarises accountability for key data activities.

Activity	Data Owner	Data Steward	Custodian	Council
Define data domain and business rules	A	R	C	I
Approve data classification	A	R	I	I
Monitor data quality	I	R	C	I
Approve access requests	A	R	C	I
Approve data disposal	A	R	R	I

R = Responsible, A = Accountable, C = Consulted, I = Informed

Appendix A: Glossary of Terms

Term	Definition
Data Lineage	The traceable record of where data originates and how it is transformed.
Dead-letter queue	A holding queue for messages that cannot be processed after retries.
Idempotency	Repeating an operation produces the same result as performing it once.
System of Record	The authoritative source system for a given data entity.